

Chi-Square Goodness of Fit Test

The Chi-Square Goodness of Fit Test is a non-parametric statistical method used to determine whether a set of observed frequencies fits a theoretical or expected distribution.

The null hypothesis H_0 assumes that the observed data follow the given theoretical distribution, while the alternative hypothesis H_1 states that there is a significant difference between the observed and expected frequencies.

The test statistic is calculated as: $\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$, where O_i and E_i represent the observed and expected frequencies for the i -th category, respectively.

Example

1. The demand for a particular spare part in a factory was found to vary from day to day. In a sample study, the following information was obtained.

Days	:	Mon	Tue	Wed	Thu	Fri	Sat
Demand	:	1124	1125	1110	1120	1126	1115

Test the hypothesis, at 5% significance level, that the number of parts demanded does not depend on the day of the week.

2. A production manager wants to check whether the number of machine breakdowns over 10 weeks follows the expected pattern predicted by the maintenance model. The observed and expected frequencies are:

Week	:	1	2	3	4	5	6	7	8	9	10
Observed (O)	:	14	18	20	17	22	19	25	21	16	13
Expected (E)	:	15	19	18	16	23	18	23	22	17	14

Test, at the 5% significance level, whether the observed breakdown frequencies fit the expected distribution.

Chi-Square Test of Independence of Attributes

The Chi-Square Test of Independence is a statistical method used to determine whether two categorical variables are independent or associated within a population. It is commonly applied when data are organized in a contingency table showing the frequency distribution across multiple categories.

The null hypothesis H_0 states that the two attributes are independent, while the alternative hypothesis H_1 states that there is a significant association between them.

The test statistic is computed as:

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}},$$

where

- O_{ij} is the observed frequency in the i -th row and j -th column,
- E_{ij} is the expected frequency, given by:

$$E_{ij} = \frac{(\text{row total}) \times (\text{column total})}{\text{grand total}}$$

The test statistic follows a Chi-Square distribution with degrees of freedom: $df = (r - 1)(c - 1)$, where r is the number of rows and c is the number of columns in the table.

The decision is based on comparing the calculated χ^2 value with the critical value from the Chi-Square distribution table at a significance level α . If the test statistic exceeds the critical value, we reject H_0 , concluding that the variables are not independent.

Example

Two sample polls of votes for two candidates A and B for a public office are taken, one from among the residents of

1. rural areas and one from urban areas. The results are given in the side table. Examine whether the nature of the area is related to voting preference in this election.

Area	Votes for		Total
	A	B	
Rural	620	380	1000
Urban	550	450	1000
Total	1170	830	2000

2. Out of 8000 graduates in a town 800 are females, out of 1600 graduate employees 120 are females. Use χ^2 to determine if any distinction is made in appointment on the basis of sex. Value of χ^2 at 5% level for one degree of freedom is 3.84.

3.

Employment Type	Prefer Remote Work	Do Not Prefer Remote Work	Total
Government	40	60	100
Private	70	30	100
Self-employed	50	50	100
Total	160	140	300

A sociologist wants to study whether the type of employment sector (Government, Private, or Self-employed) is associated with an individual's preference for remote work. A survey of 300 people is conducted and the responses are tabulated. Test at the 5% level of significance whether there is a significant association between employment type and remote work preference.

- A survey is conducted among 600 working individuals to examine whether type of employment is associated with gender.
4. The results are summarized in the table. Use the Chi-Square test at the 5% level of significance to determine whether the type of employment is independent of gender.

Gender	Government	Private	Self-Employed	Total
Male	120	150	90	360
Female	80	100	60	240
Total	200	250	150	600

5. Below is given the distribution of hair colours for either sex in a university:

Hair colour	Fair	Red	Medium	Dark	Jet black	Total
Boys	592	119	849	504	36	2100
Girls	544	97	677	451	14	1783
Total	1136	216	1526	955	50	3883

Test the homogeneity of hair colour for either sex at 5% significance level.